the use of GaAs circuitry in mobile phones, satellite communications, microwave point-to-point links, and higher frequency radar systems. Besides, GaAs is also being used to manufacture Gunn diodes for the generation of microwaves.

Next-generation GaAs semiconductors and devices promise to create a huge demand, not totally replacing the existing semiconductor market, but ultimately making a huge dent in it. Due to the mounting limitations with the use of silicon, a market of worth $500 billion dollars in 2020 is looking towards the ability to replace silicon semiconductors with more suitable semiconductors. The GaAs wafers/substrates are next-generation technology as they operate faster than the silicon semiconductors. Thus, GaAs represents the next generation of semiconductor chips, which can do things that the silicon chips cannot do.

Next-generation GaAs support the signal speed which is needed to implement 5G supported Internet of Things (IoT). Therefore, the potential for the next-generation GaAs wafers is overwhelming with the overall semiconductor market likely to surpass $20 trillion by 2026. Once world economies appreciate these developments, GaAs semiconductor markets are expected to really take off.

GaAs is a direct band-gap semiconductor with a zinc blende crystal structure and can be used for sensing (3D sensing for consumer electronics and electric vehicles). Its application in radars and for lasers is common. Further, the benefits of using GaAs in devices derive in part from the characteristic that GaAs generates less noise than most other types of semiconductor components and, as a result, it is useful in weak-signal amplification applications. Due to the benefits related to generating less noise, GaAs is again a suitable replacement for silicon in the manufacture of linear and digital integrated circuits.

Design challenges

The environment, health, and safety aspects of GaAs sources (such as trimethylgallium and arsine) and industrial hygiene monitoring studies of metalorganic precursors designate gallium arsenide as a carcinogen.

Another biggest drawback with GaAs is the high cost of a single-crystal GaAs substrate, which has been a barrier to volume manufacturing.

Further, just as in silicon devices, the seeds of GaAs contact reliability problems are often planted during processing and contribute to reducing contact quality, conducting projections that threaten to short the formed GaAs layer.

> ELECTRON MOBILITY, SINGLE JUNCTION BAND-GAP, HIGHER EFFICIENCY, HEAT AND MOISTURE RESISTANCE, AND SUPERIOR FLEXIBILITY ARE THE FIVE DISTINCT ADVANTAGES OF GAAS, WHICH ARE IMPROVING THE ACCEPTANCE OF GaAs WAFERS IN THE SEMICONDUCTOR INDUSTRY.

The application of ion implantation for the fabrication of GaAs devices has not been developed rapidly. The reasons for this lag are that the development of implantation doping techniques in GaAs matured somewhat later and thus, the development of integrated circuit technologies using GaAs have occurred only recently.

These integrated circuit technologies require the reproducibility and uniformity of doping whose implantation in case of silicon has been demonstrated. Several features of implantation in GaAs differ to that of silicon and the most notable of these is the dissociation of GaAs by loss of arsenic.

Further, service failures in GaAs devices can occur at surfaces and interfaces, in the substrate or active layers, at Schottky and ohmic contacts, and by EM and corrosion damage of the metallisation. The three main contact-degradation effects include:

- Long-term inter diffusion within metal sandwich structures
- Electric field-induced metal migration between closely spaced neighbouring electrodes
- Occurrence of metal migration through narrow conductors carrying large current densities

The performance and reliability of GaAs devices is influenced by the presence of crystal defects (dislocations, precipitates, point defects, etc) either native or produced during the technological processing. Dislocation and the crystal matrix interaction generally introduce defects forming the Cottrell atmospheres, which increase the crystal inhomogeneity at micrometric scale.

Due to low thermal conductivity (1/3rd in comparison to Si) of GaAs heat generation is concentrated in a few discrete locations, which results in large thermal gradients and non-isothermal surfaces. Therefore, thermal characterisation of GaAs devices requires detailed understanding of heat generation and temperature prediction within sub-micron length scales for two primary reasons.

The first is that electrical performance of the device is a function of temperature, and the electrical parameters of interest that vary with temperature include phase shift, output power, efficiency, and gain. The second reason to understand the thermal performance will allow the estimation of reliability. Thermal modelling is required to accurately correlate temperature-accelerated life test data because of the difficulty in measuring temperatures at such small scales.

The cooling of GaAs devices is primarily a conduction problem, which also requires an understanding of fabrication and packaging techniques as well as the device geometry. Heat flow due to convection and radiation are usually present when considering the ultimate heat sink but conduction mode of heat flow is dominant at the device level.

The basis for the silicon's superior reliability is inherent and lies in the nature of its oxide, which can be grown under controlled conditions and has better protective properties. However, the oxide of GaAs does not